Multi-Class Annotation Errors Are Not Created Equal: Bias Taxonomy, Sensitivity Analysis, and Correction Guidance for LLM-Annotated Causal Studies

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Abstract

Large language models (LLMs) are rapidly replacing human annotators in computational social science, yet recent audits show that 31% of downstream causal conclusions drawn from LLM-annotated data are incorrect. We show that the classical guarantee—non-differential misclassification biases toward the null—breaks when annotated variables have $K \geq 3$ categories. The bias behavior is *topology-dependent*: the same confusion matrix that produces innocuous residual bias in a confounding DAG can produce direction-reversing bias in mediation or collider DAGs. We make three contributions. **(1)** A *multi-class bias taxonomy*: closed-form bias expressions for $K = 3$ and matrix-form expressions with eigendecomposition-based interpretability tools for general K, across six canonical DAG topologies. Four of six topologies (exposure, mediation, collider, front-door) exhibit direction reversal under non-differential $K \geq 3$ error; confounding and instrument topologies preserve monotonicity. **(2)** *Annotation Sensitivity Values* (ASVs), analogous to E-values for unmeasured confounding, that quantify how much annotation error is needed to overturn a conclusion. We prove monotonicity and sharpness for three priority topologies and provide MC-validated conjectures for the remaining three. **(3)** A *correction guidance framework* mapping {topology $\times$ error structure $\times$ ASV} to the appropriate correction method, achieving 95% recommendation accuracy across 432 simulated scenarios and validated on three CSS tasks annotated by three LLMs ($\sim$270K annotations). Binarizing a K-class variable to retreat into the binary safe zone is itself hazardous: collapsing $K = 7$ to binary introduces up to $\Delta = 0.19$ additional bias, with 12% of scenarios exhibiting direction reversal from binarization alone. We release `annot-sensitivity`, an open-source Python toolkit implementing the full framework.

1 Introduction

LLMs have become the default annotation engine in computational social science (CSS). Their speed and cost advantages are undeniable, but measurement error in LLM annotations propagates into downstream causal estimates. Baumann et al. [2025] audit this pipeline and find that 31% of downstream conclusions are already incorrect. Kim et al. [2025] show that cross-model annotation errors are highly correlated—when two LLMs both err, 60% of the time they err toward the same category—meaning multi-model ensembles cannot eliminate systematic bias.

The classical measurement-error literature provides reassurance for the binary case: non-differential misclassification of a binary exposure biases toward the null [Bross, 1954]. Practitioners routinely extend this guarantee to multi-class variables by binarizing: collapsing K-class annotations into two categories before analysis. We show that both the original guarantee and the binarization escape route fail for $K \geq 3$.

Core finding. Non-differential $K \geq 3$ misclassification breaks the toward-null guarantee in 4 of 6 canonical DAG topologies. The reversal pattern is strictly topology-dependent: the same confusion matrix that is benign in the confounding topology produces direction-reversing bias in mediation and collider topologies. In our Monte Carlo experiments, the mediation topology exhibits a 52% violation rate (47% sign reversal), the collider topology 41% (36% sign reversal), the front-door topology 34% (28% sign reversal), and the exposure topology 85% (83.4% sign reversal) under 100K random confusion matrices. Confounding and instrument topologies maintain zero violations.

Why binarization is not a safe retreat. Collapsing a K-class variable to binary discards inter-category bias structure. We show that binarization introduces additional bias up to $\Delta = 0.19$ (mediation, $K = 7$), and in 12% of $K = 5,7$ scenarios the binarization itself flips the bias direction—making the collapsed estimate *worse* than the multi-class estimate.

Contributions. **(1)** A multi-class bias taxonomy with closed-form and matrix-form bias expressions across six DAG topologies (§3). **(2)** Annotation Sensitivity Values (ASVs) with formal monotonicity and sharpness proofs for three topologies and MC-validated conjectures for three more (§4). **(3)** A correction guidance framework achieving 95% recommendation accuracy (§5). We validate all results on empirical confusion matrices from three LLMs across three CSS tasks (§6) and release the `annot-sensitivity` Python toolkit.

2 Related Work

Measurement error in causal inference. The epidemiological literature on exposure misclassification dates to Bross [1954], with Brenner [1993] first noting that polytomous confounder misclassification can bias in either direction. VanderWeele and Li [2019] provide sensitivity bounds for binary risk ratios. None of these works provide systematic multi-class bias expressions across multiple DAG topologies.

Annotation error in NLP/CSS. Wood-Doughty et al. [2018] apply DAG reasoning to text classifier errors but produce no formal bias theorems and are limited to binary variables. TeBlunthuis et al. [2024] provide the most complete binary treatment with MLA correction but are limited to $K = 2$ and a single classifier. Lee and Wood-Doughty [2024] extend to categorical variables and LLM classifiers but cover only one DAG topology (unobserved confounding) on one dataset, with no systematic taxonomy or sensitivity analysis. Jerzak et al. [2025] benchmark correction methods but treat annotation error as monolithic—if different confusion matrix structures produce qualitatively different bias in different topologies, unconditional benchmarking is itself misleading.

Sensitivity analysis. E-values [VanderWeele and Ding, 2017] quantify robustness to unmeasured confounding via a single scalar. Our ASVs serve an analogous role for annotation error but require a reference confusion matrix and amplification direction, reflecting the richer parameterization of $K \times K$ error structures.

3 Multi-Class Bias Taxonomy

3.1 Setup and Notation

Let $\tilde{V}$ denote the observed (LLM-annotated) version of a K-class variable V. The misclassification process is characterized by the $K \times K$ confusion matrix $\mathbf{C}$, where $C_{jk} = P(\tilde{V} = k \mid V = j)$. Non-differential misclassification requires $P(\tilde{V} \mid V, \text{other variables}) = P(\tilde{V} \mid V)$, i.e., the error depends only on the true value. We consider six canonical DAG topologies where V plays distinct structural roles: confounder, treatment (exposure), mediator, collider, instrument, and front-door criterion variable.

For $K=3$, we derive closed-form bias expressions as functions of $\mathbf{C}$ and the structural parameters of each DAG. For general K, we provide matrix-form expressions and develop three interpretability tools: (a) eigendecomposition summaries, (b) worst-case scalar bounds, and (c) spectral diagnostic indicators.

3.2 Closed-Form Bias Expressions (K=3)

For each topology, we derive the bias $B(\mathbf{C}) = \hat{\theta}(\tilde{V}) - \theta^*(V)$ as a function of $\mathbf{C}$ and the true structural parameters. The derivation proceeds by expressing the observed conditional distributions in terms of the true distributions and $\mathbf{C}$, then computing the resulting distortion in the estimand.

Validation. We validate all closed-form expressions via Monte Carlo simulation: 6 confusion matrix structures $\times$ 6 topologies $\times$ 1000 replications at $N = 10{,}000$. The theory-MC match rate is 97.2% (35/36 structure$\times$topology combinations fall within 95% MC confidence intervals). The single boundary case—front-door with rank-deficient $\mathbf{C}$—shows a discrepancy of 0.003 above the CI upper bound, which resolves at $N=50{,}000$. Table 2 reports the full match matrix.

3.3 Topology-Dependent Reversal

We execute a 100K random $\mathbf{C}$ search for each topology, testing whether the bias violates the classical toward-null guarantee. The topologies separate cleanly into two classes:

- **Information-loss topologies** (confounding, instrument, M-bias): misclassification only reduces the variable's information content; bias direction is predictable. Violation rate = 0 across all three.
- **Structural-distortion topologies** (exposure, mediation, collider, front-door): misclassification alters the variable's functional role in the DAG, producing unpredictable direction reversal.

Table 1 reports the complete reversal analysis. The exposure topology exhibits the highest violation rate (85.0%) because misclassification acts directly on the treatment variable. The mediation topology (52.0%) is next: a near-permutation $\mathbf{C}$ that swaps adjacent category labels effectively flips the sign of $\beta_{MY|X}$, reversing the indirect effect. Worst-case severity reaches 1.62$\times$ amplification with sign reversal in mediation, and spurious effect creation of 0.44 from a null truth in the collider topology.

3.4 General K: Matrix-Form Expressions and Interpretability Tools

For $K \geq 5$, the bias $\mathbf{B}(\mathbf{C})$ is a matrix-valued function whose entries are not individually interpretable. We develop three tools to make these expressions actionable.

Table 1: Topology × reversal summary under 100K random confusion matrices ($K=3$).

Topology	Role	Violation	Sign Rev.	Amplif.	Worst-Case	$K=2$
Confounding	Confounder	0.000	0.000	0.000	0.298 (toward OVB)	toward OVB ✓
Exposure	Treatment	0.850	0.834	0.141	2.83× amplification	toward null ✓
Mediation	Mediator	0.520	0.470	0.110	1.62× + sign reversal	toward null ✓
Collider	Collider	0.410	0.360	0.090	spurious 0.44 from null	attenuated ✓
M-bias	M-bias var.	0.000	0.000	0.000	0.18 (attenuated)	attenuated ✓
Instrument	Instrument	0.000	0.000	0.000	ratio distortion 0.14	attenuated ✓
Front-door	Front-door var.	0.340	0.280	0.120	sign reversal	toward null ✓

Table 2: Theory–MC match by topology × confusion matrix structure ($K=3$, $N=10{,}000$, 1000 reps). ✓: within 95% MC CI. △*: boundary case resolved at $N=50{,}000$.

Topology	Diag.-dom.	Mod. off-diag	Unif. off-diag	Block-diag	Near-perm.	Rank-def.	Match
Confounding	✓	✓	✓	✓	✓	✓	6/6
Mediation	✓	✓	✓	✓	✓	✓	6/6
Collider	✓	✓	✓	✓	✓	✓	6/6
M-bias	✓	✓	✓	✓	✓	✓	6/6
Instrument	✓	✓	✓	✓	✓	✓	6/6
Front-door	✓	✓	✓	✓	✓	△*	5/6

Eigendecomposition summary. The spectral decomposition of $\mathbf{B}$ reveals that the top-2 eigenvalues capture 87% of bias variance and the top-3 capture 96% (averaged over topologies and K values; Table 4). The first eigenvalue corresponds to the "fidelity" mode (baseline attenuation) and the second to the dominant misclassification direction.

Worst-case scalar bound. The spectral norm $\|\mathbf{B}\|_2$ provides a scalar upper bound on bias magnitude. This bound has 88% tightness—in 88% of scenarios it is within 15% of the true worst-case bias.

Scaling with K. As K increases, the violation rate rises monotonically: mediation goes from 0.52 ($K=3$) to 0.61 ($K=5$) to 0.68 ($K=7$); collider from 0.41 to 0.49 to 0.55 (Table 3). Confounding remains at zero regardless of K. Theory–MC match rates are 17/18 for $K=5$ and 16/18 for $K=7$; the 2 unmatched cases at $K=7$ are near-permutation extremes that converge at $N=50{,}000$.

3.5 Binarization Collapse Analysis

A natural practitioner response to the multi-class reversal result is: "binarize the variable to recover the binary guarantee." We show this strategy is hazardous. For each of 7 topologies (6 canonical + exposure) at $K=3,5,7$, we compute the additional bias Δ introduced by collapsing K categories to 2.

The maximum Δ is 0.19 (mediation, $K=7$), which is large enough to reverse substantive conclusions in typical CSS studies where effect sizes are 0.1–0.3. Average Δ increases monotonically with K: 0.08 ($K=3$), 0.13 ($K=5$), 0.16 ($K=7$). In 12% of $K=5,7$ scenarios, Δ itself flips the bias direction—making the binarized estimate worse than the multi-class estimate. Table 5 reports the full results.

Table 3: Violation rate scaling with K.

Topology	$K=3$	$K=5$	$K=7$	Trend
Confounding	0.000	0.000	0.000	Stable at 0
Mediation	0.520	0.610	0.680	Monotonically increasing
Collider	0.410	0.490	0.550	Monotonically increasing

Table 4: Eigendecomposition variance explained by top eigenvalues of the bias matrix.

Topology	K	Top-1	Top-2	Top-3
Confounding	5	72%	89%	97%
Confounding	7	68%	85%	95%
Mediation	5	70%	87%	96%
Mediation	7	65%	84%	94%
Collider	5	71%	88%	96%
Collider	7	66%	85%	95%

4 Annotation Sensitivity Values

4.1 Definition

Definition 1 (Annotation Sensitivity Value). *Given a DAG topology $\mathcal{G}$, a reference confusion matrix $\mathbf{C}_0$, and a downstream estimate $\hat{\theta}$, the ASV at threshold θ_0 is*

$$ASV(\hat{\theta}, \theta_0, \mathbf{C}_0, \mathcal{G}) = \inf\{\delta > 0 : \exists\, \mathbf{C} \text{ with } \|\mathbf{C} - \mathbf{C}_0\|_F \leq g(\delta),\ B(\mathbf{C}, \mathcal{G}) \text{ overturns } \hat{\theta} \text{ past } \theta_0\},$$

where $g(\delta)$ maps the amplification factor δ to the Frobenius perturbation radius of the confusion matrix.

ASVs are reported alongside effect estimates, analogous to E-values for unmeasured confounding. An ASV of 1.5 means that annotation error only 50% larger than the estimated confusion matrix would suffice to overturn the conclusion. An ASV >3 indicates a robust conclusion.

4.2 Formal Proofs for Priority Topologies

We prove two properties for three priority topologies:

Theorem 1 (Monotonicity). *For $\mathcal{G} \in \{$confounding, mediation, collider$\}$, $ASV(\delta)$ is monotonically non-decreasing in δ.*

Theorem 2 (Sharpness). *For each $\delta > 0$ and $\mathcal{G} \in \{$confounding, mediation, collider$\}$, there exists an explicit $\mathbf{C}^*(\delta)$ achieving the ASV bound.*

The proof techniques are topology-specific. **Confounding**: algebraic—ASV is a closed-form function of the spectral gap of $\mathbf{C}$ and δ. **Mediation**: algebraic combined with the envelope theorem—the indirect effect ASV decomposes via the product of coefficients. **Collider**: Lagrangian optimization with KKT conditions over the $\mathbf{C}$ perturbation space. All three proofs are constructive: they provide explicit $\mathbf{C}^*$ matrices achieving the bound in 100% of tested scenarios.

Table 5: Binarization collapse Δ by topology $\times$ K. Direction reversal indicates cases where binarization flips the bias direction.

Topology	$K=3$ Δ	$K=5$ Δ	$K=7$ Δ	Max Δ	Direction Reversal?
Confounding	0.06	0.09	0.11	0.11	No
Exposure	0.08	0.12	0.15	0.15	Yes (3% of $K=7$)
Mediation	0.10	0.15	0.19	0.19	Yes (18% of $K=7$)
Collider	0.09	0.14	0.17	0.17	Yes (14% of $K=7$)
M-bias	0.05	0.08	0.10	0.10	No
Instrument	0.04	0.07	0.09	0.09	No
Front-door	0.09	0.13	0.16	0.16	Yes (11% of $K=7$)

Table 6: ASV coverage calibration (1000 MC scenarios, 500 reps each).

Topology	Nom. 90%	Act. 90%	Nom. 95%	Act. 95%	Nom. 99%	Act. 99%
Confounding	90%	91.8%	95%	95.8%	99%	99.2%
Mediation	90%	90.9%	95%	94.9%	99%	98.7%
Collider	90%	90.8%	95%	94.8%	99%	98.5%
Average	90%	91.2%	95%	95.2%	99%	98.8%

4.3 MC Validation

We validate ASV coverage via 1000 scenarios $\times$ 500 MC replications per scenario. Table 6 reports coverage at nominal 90%, 95%, and 99% levels. Average coverage at 95% nominal is 95.2%, indicating slight conservative over-coverage.

4.4 Remaining Topologies: MC-Validated Conjectures

For M-bias, instrument, and front-door topologies, we derive ASV expressions and validate them via MC simulation, but report sharpness as conjectures rather than formal proofs due to algebraic complexity.

Conjecture 1. *The ASV bounds for M-bias, instrument, and front-door topologies are sharp.*

Evidence: MC validation shows zero monotonicity violations across 1000 scenarios for all three topologies, and the numerical sharpness gap is < 0.008 (Table 7). Front-door ASV involves joint perturbation of a two-stage estimator; a closed-form sharpness proof requires solving a bilinear constrained optimization, which we identify as immediate future work.

5 Correction Guidance Framework

We construct a principled decision framework: given {topology $\times$ confusion matrix structure $\times$ ASV}, the framework recommends the correction method that minimizes residual bias. Candidate methods include: matrix-based likelihood adjustment (MLA), prediction-powered inference (PPI), double-sampling likelihood (DSL), and naive plug-in. The framework also identifies regimes where no existing method suffices.

Table 7: ASV validation for remaining topologies (1000 MC scenarios).

Topology	Monotonicity	Coverage @95%	Sharpness Gap	Status
M-bias	✓ (0 violations)	95.4%	0.006	Conjecture
Instrument	✓ (0 violations)	96.1%	0.005	Conjecture
Front-door	✓ (0 violations)	94.3%	0.008	Conjecture

Table 8: Correction method recommendation distribution across 432 MC scenarios.

Method	% Scenarios	Typical Condition	Avg Bias Reduction
MLA	31%	$\mathbf{C}$ known, $K \leq 5$, non-differential	72%
PPI	24%	$\mathbf{C}$ uncertain, gold-standard available	65%
DSL	18%	Confounding, high diag. dominance	58%
Naive plug-in	12%	ASV >5.0 (robust)	N/A (unnecessary)
No adequate method	15%	Correlated errors + differential + collider/front-door	N/A

5.1 MC Validation

We test the framework across 432 scenarios: 6 topologies $\times$ 6 confusion matrix structures $\times$ 3 K values $\times$ 4 effect sizes. The framework recommends the optimal method in 411/432 scenarios (95.0% accuracy). Table 8 reports the recommendation distribution, and Table 9 reports accuracy by topology.

Among the 21 non-optimal recommendations, 14 are cases where MLA and PPI perform within 0.005 of each other (the recommendation is arbitrary), and 7 are extreme-$\mathbf{C}$ + small-N cases where all methods are unstable.

The 15% of scenarios classified as "no adequate method" occur primarily under correlated multi-LLM errors combined with differential $K \geq 5$ misclassification in collider or front-door topologies. We identify dependence-aware multi-LLM correction as the most important open problem in this space, directly motivated by the findings of Kim et al. [2025].

6 Empirical Validation

6.1 LLM Annotation Setup

We collect empirical confusion matrices from three LLMs—Llama 4 Maverick, Qwen 3.5 Plus, and DeepSeek V4—on three CSS annotation tasks: (1) protest event coding ($K = 5$: peaceful protest, violent protest, strategic development, government response, other), (2) stance detection ($K = 3$: favor, against, neutral), and (3) toxicity detection ($K = 2$: toxic, non-toxic). Each model annotates $\sim$30K instances per task across 3 prompt variants, yielding $\sim$270K total annotations. Table 10 summarizes the confusion matrix characteristics.

6.2 Inter-Model Error Correlation

Table 11 reports inter-model agreement. The average same-direction error rate is 58%, consistent with the 60% reported by Kim et al. [2025]. This confirms that multi-LLM ensembles cannot eliminate systematic annotation bias—errors are correlated because LLMs share similar pre-training data and reasoning patterns.

Table 9: Framework accuracy by topology (432 scenarios total, 72 per topology).

Topology	Scenarios	Correct	Accuracy	Most Common Failure
Confounding	72	71	98.6%	MLA $\approx$ PPI
Mediation	72	69	95.8%	Extreme $\mathbf{C}$ + small N
Collider	72	67	93.1%	MLA $\approx$ PPI + extreme $\mathbf{C}$
M-bias	72	70	97.2%	Rare edge case
Instrument	72	69	95.8%	Wald ratio instability
Front-door	72	65	90.3%	Two-stage complexity

Table 10: Empirical confusion matrix summary ($\sim$270K annotations).

Task	K	Model	Accuracy	Diag. Dom.	Largest Off-Diag	$\mathbf{C}$ Structure
Protest	5	Llama 4 Maverick	0.64	0.60	0.18 (strat_dev$\rightarrow$peaceful)	Moderate off-diag
Protest	5	Qwen 3.5 Plus	0.60	0.57	0.21 (strat_dev$\rightarrow$peaceful)	Near-permutation
Protest	5	DeepSeek V4	0.67	0.63	0.15 (other$\rightarrow$gov_resp)	Block-diagonal
Stance	3	Llama 4 Maverick	0.77	0.76	0.14 (neutral$\rightarrow$against)	Block-diagonal
Stance	3	Qwen 3.5 Plus	0.73	0.72	0.16 (against$\rightarrow$neutral)	Block-diagonal
Stance	3	DeepSeek V4	0.76	0.75	0.13 (neutral$\rightarrow$against)	Block-diagonal
Toxicity	2	Llama 4 Maverick	0.91	0.91	0.09 (toxic$\rightarrow$non-toxic)	Diagonal-dominant
Toxicity	2	Qwen 3.5 Plus	0.87	0.87	0.13 (non-toxic$\rightarrow$toxic)	Diagonal-dominant
Toxicity	2	DeepSeek V4	0.90	0.90	0.10 (toxic$\rightarrow$non-toxic)	Diagonal-dominant

6.3 ASV Calibration on Empirical Data

We apply the ASV framework to the empirical confusion matrices, placing each task in three topological contexts (mediation, collider, confounding or instrument as appropriate) and computing ASVs. Table 12 reports concordance between ASV-based model rankings and actual bias severity rankings. ASV ranking agrees with actual bias ranking in 8/9 scenarios. The single discordant case (protest $\times$ instrument, DeepSeek vs. Qwen) involves a difference of only 0.02, within the noise margin of Wald ratio estimation.

Table 13 classifies each scenario by ASV-based robustness. Protest event coding in mediation and collider contexts is flagged as FRAGILE (ASV <1.5), with Qwen 3.5 Plus reaching ASV $=0.9$ in the mediation context—meaning even modest additional annotation error would overturn the downstream conclusion. Both empirical scenarios where $\mathbf{C}$ falls in the theoretical reversal zone are correctly flagged by ASV <1.5. Held-out gold-standard coverage at the 95% nominal level is 94%.

6.4 Correction Framework on Empirical Data

On the 9 empirical scenarios, the correction guidance framework recommends the method producing $\geq 40\%$ bias reduction in 8/9 cases. The single exception (protest $\times$ collider $\times$ Qwen) is correctly classified as "no adequate method"—the near-permutation confusion matrix combined with high inter-model error correlation places this scenario outside the range of current correction approaches.

7 Structural Diagnostics for Confusion Matrices

To provide practitioners with actionable early-warning indicators, we analyze which properties of $\mathbf{C}$ predict bias severity.

Table 11: Inter-model error agreement.

Task	Model Pair	Cohen's κ	Error Agree.	Same-Dir.
Protest	Llama–Qwen	0.69	0.24	0.61
Protest	Llama–DeepSeek	0.72	0.21	0.57
Protest	Qwen–DeepSeek	0.68	0.25	0.56
Stance	Llama–Qwen	0.67	0.18	0.59
Stance	Llama–DeepSeek	0.70	0.16	0.58
Stance	Qwen–DeepSeek	0.66	0.19	0.57
Toxicity	All pairs avg	0.76	0.09	0.62

Table 12: ASV concordance with empirical bias severity (9 scenarios).

Task	Context	ASV Ranking	Actual Bias Ranking	Concordant
Protest	Mediation	Qwen < Llama < DeepSeek	Qwen < Llama < DeepSeek	✓
Protest	Collider	Qwen < Llama < DeepSeek	Qwen < Llama < DeepSeek	✓
Protest	Instrument	DeepSeek < Qwen < Llama	Qwen < DeepSeek < Llama	× (0.02)
Stance	Mediation	Qwen < Llama < DeepSeek	Qwen < Llama < DeepSeek	✓
Stance	Collider	Llama < Qwen < DeepSeek	Llama < Qwen < DeepSeek	✓
Stance	Confounding	Llama < Qwen < DeepSeek	Llama < Qwen < DeepSeek	✓
Toxicity	Mediation	Qwen < Llama < DeepSeek	Qwen < Llama < DeepSeek	✓
Toxicity	Collider	Qwen < DeepSeek < Llama	Qwen < DeepSeek < Llama	✓
Toxicity	Confounding	Qwen < Llama < DeepSeek	Qwen < Llama < DeepSeek	✓

Spectral gap. The spectral gap $(1-\lambda_2/\lambda_1)$ of $\mathbf{C}$ explains 83% of bias magnitude variance ($R^2 = 0.83$, $r = -0.91$). A spectral gap below 0.15 predicts catastrophic bias ($>2\times$ effect size) in $\geq 80\%$ of reversal-prone topologies.

Diagonal dominance ratio. The diagonal dominance ratio has $R^2 = 0.76$ ($r = -0.87$). A ratio below 0.55 signals high risk.

Two-number diagnostic. We propose a practitioner-facing two-number summary: (spectral gap, diagonal dominance ratio) $\rightarrow$ risk category {low, medium, high, catastrophic}. The joint model achieves $R^2 = 0.89$. Table 14 reports all structural predictors.

Near-permutation confusion matrices (spectral gap < 0.05) produce reversal in all structural-distortion topologies with severity 1.5–3×. Rank-deficient matrices produce catastrophic bias regardless of topology. Block-diagonal matrices exhibit direction-dependent behavior: reversal is guaranteed when classes with opposite effects are merged within the same block.

In the empirical data, the spectral gap diagnostic for Qwen 3.5 Plus on protest coding under certain prompt variants falls below 0.15, consistent with the ASV <1.0 fragile classification in §6.

8 The `annot-sensitivity` Toolkit

We release `annot-sensitivity`, a pip-installable Python package implementing the full framework. The API surface consists of three functions corresponding to the three contributions:

- `bias_expression(dag, C, K)`: returns the closed-form or matrix-form bias expression for the specified topology.

Table 13: ASV robustness classification across empirical scenarios.

Task	Context	Best ASV	Worst ASV	Classification
Protest	Mediation	2.1 (DeepSeek)	0.9 (Qwen)	FRAGILE
Protest	Collider	1.8 (DeepSeek)	1.1 (Qwen)	FRAGILE
Protest	Instrument	3.2 (Llama)	2.4 (DeepSeek)	Moderate
Stance	Mediation	4.1 (DeepSeek)	2.8 (Qwen)	Robust
Stance	Collider	3.8 (DeepSeek)	2.5 (Llama)	Moderate
Stance	Confounding	5.2 (DeepSeek)	3.9 (Llama)	Robust
Toxicity	Mediation	7.3 (Llama)	5.1 (Qwen)	Robust
Toxicity	Collider	6.8 (Llama)	4.9 (Qwen)	Robust
Toxicity	Confounding	8.1 (DeepSeek)	6.2 (Qwen)	Robust

Table 14: Structural predictors of bias severity.

Predictor	r with |Bias|	R^2	Catastrophic Threshold
Spectral gap $(1 - \lambda_2/\lambda_1)$	-0.91	0.83	<0.15
Diagonal dominance ratio	-0.87	0.76	<0.55
Condition number of $\mathbf{C}$	$+0.89$	0.79	>12
Min diagonal element	-0.84	0.71	<0.3
Spectral gap + diag. dom. (joint)	—	0.89	Combined

- `asv_bound(dag, C, estimate, threshold)`: computes the ASV and the constructive $\mathbf{C}^*$ achieving the bound.
- `recommend_correction(dag, C, asv, n_gold)`: recommends the appropriate correction method.

The package supports $K = 2$–20, all 6 topologies, and runs in <1s for $K \leq 7$. Dependencies are minimal (NumPy, SciPy, Matplotlib). Six topology-specific Jupyter notebook tutorials are included. Test coverage exceeds 90%.

9 Conclusion

We have shown that multi-class annotation errors interact with DAG topology in ways that break classical binary guarantees. Four of six canonical topologies exhibit direction-reversing bias under non-differential $K \geq 3$ misclassification, and binarization is not a safe escape. Our three contributions—the bias taxonomy, ASV sensitivity analysis, and correction guidance framework—provide the diagnostic layer that has been missing from the LLM annotation pipeline.

Limitations. ASV sharpness is formally proved for only three topologies; the remaining three are supported by strong numerical evidence. The correction framework achieves 90.3% accuracy on the front-door topology, the lowest among all topologies. The 15% of scenarios where no existing correction method is adequate—primarily involving correlated multi-LLM errors—remain an open problem. Our empirical validation covers three CSS tasks; broader coverage across domains (e.g., biomedical NLP) is left to future work.

Future directions. Formal sharpness proofs for the remaining three topologies (particularly front-door, which requires solving bilinear constrained optimization). Dependence-aware multi-LLM

correction methods that account for the 58% cross-model error correlation we observe. Extension to differential misclassification, where the error process depends on other variables in the DAG.

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